MODEL LAB · 2026-10-11

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

29,344
held-out plate appearances

3.06%
held-out HR rate

0.5992
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-08-31, scored on the 29,344 that came after (actual HR rate 3.06%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5832	1.56×	0.02964	0.13599	0.00484
KNN comps Prior is the pooled rate of the nearest comparables.	0.5992	1.54×	0.02960	0.13502	0.00474
SVR Support-vector regression on the same feature matrix.	0.5923	1.41×	0.02964	0.13589	0.00646
Random forest Forest regression; a learned similarity metric.	0.5917	1.41×	0.02963	0.13553	0.00590
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5992	1.54×	0.02960	0.13502	0.00474
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5989	1.52×	0.02960	0.13504	0.00512
Linear regression Ordinary least squares on the same features.	0.5936	1.48×	0.02962	0.13560	0.00490
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5949	1.57×	0.02963	0.13552	0.00586
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5937	1.50×	0.02962	0.13544	0.00581
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5926	1.41×	0.02964	0.13580	0.00659

BLEND WEIGHTS (FOREST VARIANT)

KNN **0.90** SVR **0.00** RF **0.10**

SHRINKAGE K PER PRIOR

league **251** knn **251** svr **251** rf **251** core **251** ensemble **251** linear **251** logistic **251** nn **251** xgb **251**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,195 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5895	1.54×	0.02382	0.11470	0.01190
KNN comps Prior is the pooled rate of the nearest comparables.	0.5767	1.28×	0.02386	0.11508	0.01196
SVR Support-vector regression on the same feature matrix.	0.5606	1.15×	0.02402	0.11973	0.01671
Random forest Forest regression; a learned similarity metric.	0.5694	1.15×	0.02396	0.11638	0.01583
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5767	1.28×	0.02386	0.11508	0.01196
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5783	1.28×	0.02386	0.11509	0.01347
Linear regression Ordinary least squares on the same features.	0.5764	1.02×	0.02394	0.11808	0.01456
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5868	1.41×	0.02390	0.11586	0.01389
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5765	1.41×	0.02390	0.11596	0.01381
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5738	1.15×	0.02402	0.11848	0.01664

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Shohei Ohtani 15.9%, Jackson Chourio 15.2%, William Contreras 12.7%, Jake Bauers 12.6%, Teoscar Hernández 12.6%, Garrett Mitchell 12.4%
Flat league	5/6	Jackson Chourio 15.1%, Shohei Ohtani 14.4%, William Contreras 12.6%, Will Smith 11.9%, Teoscar Hernández 11.9%, Jake Bauers 11.8%
KNN comps	6/6	Shohei Ohtani 15.9%, Jackson Chourio 15.4%, Jake Bauers 12.8%, William Contreras 12.7%, Teoscar Hernández 12.6%, Garrett Mitchell 12.1%
SVR	5/6	Jackson Chourio 15.8%, Shohei Ohtani 15.8%, Jake Bauers 12.8%, Teoscar Hernández 12.7%, William Contreras 12.5%, Will Smith 11.9%
Random forest	5/6	Shohei Ohtani 16.2%, Jackson Chourio 15.9%, Teoscar Hernández 12.9%, Jake Bauers 12.7%, William Contreras 12.6%, Will Smith 12.5%
Core (KNN+SVR)	6/6	Shohei Ohtani 15.9%, Jackson Chourio 15.4%, Jake Bauers 12.8%, William Contreras 12.7%, Teoscar Hernández 12.6%, Garrett Mitchell 12.1%
Forest blend	5/6	Shohei Ohtani 15.9%, Jackson Chourio 15.4%, Jake Bauers 12.8%, Teoscar Hernández 12.7%, William Contreras 12.7%, Will Smith 12.1%
Linear regression	5/6	Shohei Ohtani 15.9%, Jackson Chourio 15.6%, Teoscar Hernández 12.8%, Will Smith 12.7%, Jake Bauers 12.7%, William Contreras 12.6%
Logistic regression	5/6	Shohei Ohtani 16.3%, Jackson Chourio 15.7%, Teoscar Hernández 12.9%, William Contreras 12.7%, Jake Bauers 12.6%, Will Smith 12.5%
Neural net (torch)	5/6	Shohei Ohtani 16.2%, Jackson Chourio 15.7%, Teoscar Hernández 12.9%, Jake Bauers 12.7%, William Contreras 12.7%, Will Smith 12.6%
XGBoost	5/6	Shohei Ohtani 16.5%, Jackson Chourio 15.9%, Jake Bauers 13.0%, Teoscar Hernández 12.9%, William Contreras 12.7%, Will Smith 12.3%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Shohei Ohtani**	LAD	**15.9%**	14.4%	15.9%	15.8%	16.2%	15.9%	15.9%	15.9%	16.3%	
2	**Jackson Chourio**	MIL	**15.2%**	15.1%	15.4%	15.8%	15.9%	15.4%	15.4%	15.6%	15.7%	
3	**William Contreras**	MIL	**12.7%**	12.6%	12.7%	12.5%	12.6%	12.7%	12.7%	12.6%	12.7%	
4	**Jake Bauers**	MIL	**12.6%**	11.8%	12.8%	12.8%	12.7%	12.8%	12.8%	12.7%	12.6%	
5	**Teoscar Hernández**	LAD	**12.6%**	11.9%	12.6%	12.7%	12.9%	12.6%	12.7%	12.8%	12.9%	
6	**Garrett Mitchell**	MIL	**12.4%**	11.7%	12.1%	11.0%	11.4%	12.1%	12.0%	11.6%	11.5%	
7	**Will Smith**	LAD	**12.2%**	11.9%	12.0%	11.9%	12.5%	12.0%	12.1%	12.7%	12.5%	
8	**Freddie Freeman**	LAD	**11.6%**	11.3%	11.4%	11.3%	11.2%	11.4%	11.4%	11.6%	11.4%	
9	**Mookie Betts**	LAD	**11.3%**	11.3%	11.1%	11.6%	11.2%	11.1%	11.1%	11.5%	11.5%	
10	**Brice Turang**	MIL	**10.9%**	11.4%	11.0%	11.1%	10.9%	11.0%	11.0%	10.5%	10.4%	
11	**Kyle Tucker**	LAD	**10.6%**	10.8%	10.5%	10.6%	10.6%	10.5%	10.5%	10.6%	10.3%	
12	**Christian Yelich**	MIL	**10.4%**	10.6%	10.8%	10.0%	10.6%	10.8%	10.8%	11.0%	10.4%	
13	**Andy Pages**	LAD	**9.4%**	8.9%	9.1%	8.9%	9.0%	9.1%	9.1%	8.8%	8.7%	
14	**Enrique Hernández**	LAD	**8.5%**	8.3%	8.2%	8.0%	8.3%	8.2%	8.2%	8.4%	8.0%	
15	**Tommy Edman**	LAD	**6.0%**	6.6%	6.0%	6.0%	5.6%	6.0%	5.9%	6.1%	6.0%	
16	**Joey Ortiz**	MIL	**5.8%**	6.9%	6.0%	6.0%	6.1%	6.0%	6.0%	5.9%	6.0%	
17	**Cooper Pratt**	MIL	**5.7%**	7.4%	5.8%	5.1%	5.6%	5.8%	5.8%	5.0%	5.2%	
18	**Sal Frelick**	MIL	**5.3%**	6.8%	5.5%	5.3%	5.0%	5.5%	5.5%	5.2%	5.3%	
19	**David Hamilton**	MIL	**5.1%**	6.5%	4.9%	4.9%	4.8%	4.9%	4.8%	5.1%	5.0%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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